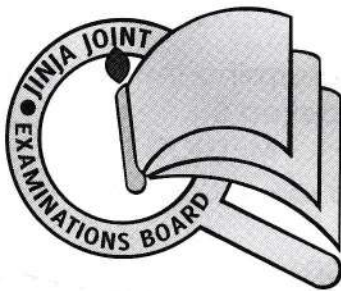


P510/3
PRACTICAL
PHYSICS
Paper 3
August, 2018
3¼ hours



JINJA JOINT EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

MOCK EXAMINATIONS – AUGUST, 2018

PHYSICS PRACTICAL

(PRINCIPAL SUBJECT)

Paper 3

3 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

*Answer **Question 1** and **one** other question.*

*Candidates are not allowed to use the apparatus or write for the first fifteen minutes.
Graph papers are provided.*

*Mathematical tables and non – programmable silent electronic calculators may be used.
Write on one side of the paper only.*

*Candidates are expected to record on their scripts all their observations as these
observations are made and to plan the presentation of the records so that it is not
necessary to make a fair copy of them. The working of the answers is to be handed in.*

*Details on the question paper should **not** be repeated in the answer, nor is the theory of
the experiment required unless specifically asked for.*

*Candidates should, however, record any special precautions that they have taken and
any particular features of their method of going about the experiment.*

*Marks are given mainly for a clear record of the observations actually made, for their
suitability and accuracy, and for the use made of them.*

1. In this experiment, you will determine the Young's modulus, E , of the material of the metre rule labelled B.

METHOD I

- a) Measure and record the thickness, t , and the breadth, w , of the metre rule labelled B.
b) Calculate k from the expression

$$k = 4wt^3$$

- c) Clamp the metre rule B at the edge of the table leaving a free length, $x = 0.900$ m as in the Figure 1.1.

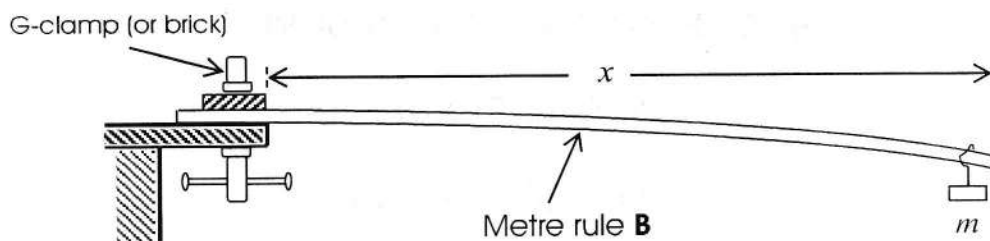


Fig. 1.1

- d) Suspend a mass, $m = 0.100$ kg at the free end of the metre rule.
e) Depress the mass, m , through a small distance and release it.
f) Measure and record the time for 20 oscillations and determine the period, T .
g) Calculate Young's modulus, E , from the expression

$$E = 6.32 \times 10^2 \left(\frac{mx^3}{kT^2} \right)$$

METHOD II

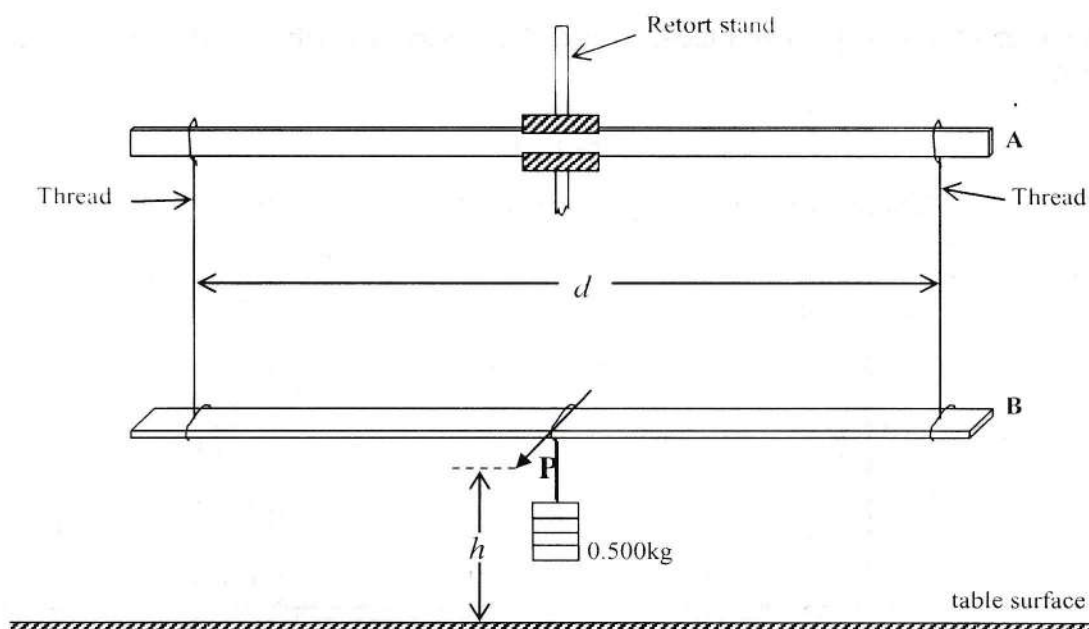


Fig. 1.2

- Clamp the metre rule labelled **A** horizontally such that the graduated face is towards you.
- Suspend metre rule **B** from the clamped one by means of two vertical threads tied at the 2.0 cm and 98.0 cm marks so that the graduated face is upwards as in the Figure 1.2.
- Adjust the length of the threads to about 20.0 cm.
- At the 50.0 cm mark of metre rule **B**, suspend a mass of 0.500 kg and attach a pointer, **P**, horizontally.
- Measure and record the initial height, h_0 , (in metres) of the pointer from the table surface.
- Remove a mass $m = 0.150$ kg from the mass hanger.
- Measure and record the new height, h_1 , (in metres) of the pointer from the table surface.
- Find the change, $y = (h_1 - h_0)$ in the position of the pointer.
- Repeat procedures (f) to (h) for $m = 0.200, 0.250, 0.300, 0.350$ and 0.400 kg.
- Tabulate your results including values of $\frac{1}{m}$ and $\frac{1}{y}$.
- Plot a graph of $\frac{1}{y}$ against $\frac{1}{m}$.
- Find the slope, S , of the graph.
- Calculate Young's modulus, E , from the expression

$$E = \frac{gSd^3}{k}$$

where $g = 9.81 \text{ ms}^{-2}$.

- Calculate the average value of E obtained by the two methods.

2. In this experiment, you will determine the focal length, f , of the converging lens provided.

PART I

a) Arrange your apparatus in a straight line as shown in Figure 2.1.

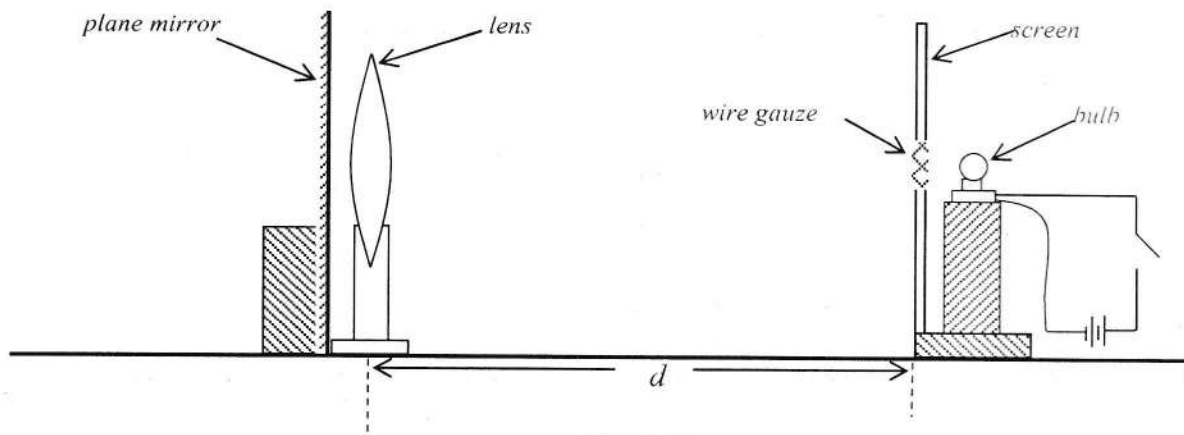


Fig. 2.1

b) Close the switch and move the lens together with the mirror to and fro the screen until a sharp distinct image of the wire gauze is formed beside the wire gauze on the screen.

c) Measure and record the distance, d , between the screen and the lens.

PART II

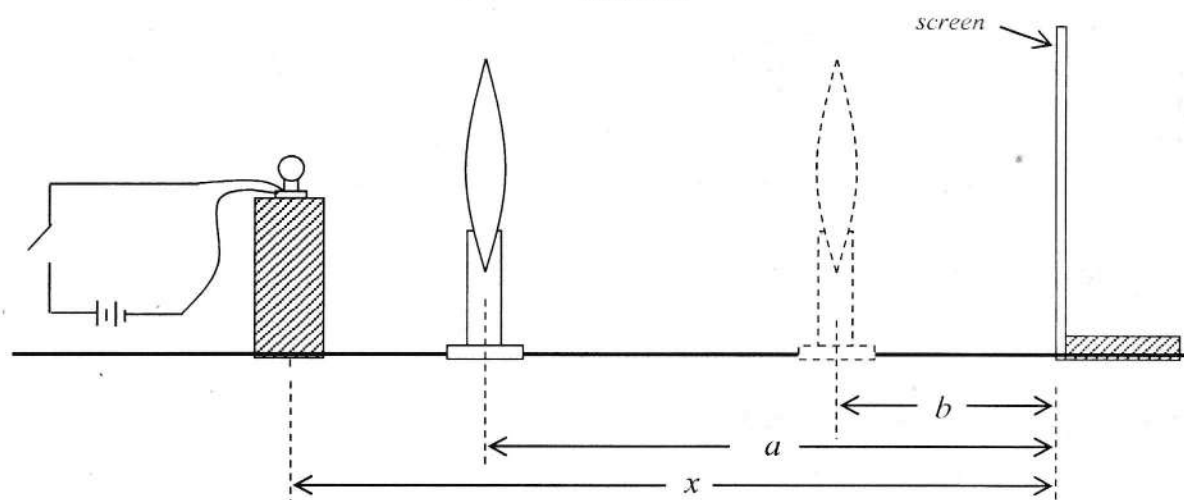


Fig. 2.2

a) Arrange the apparatus in a straight line as shown in Figure 2.2 with the distance, x , between the bulb and the screen equal to 95.0 cm.

- b) Close the switch and move the lens starting from near the bulb towards the screen until a sharp magnified image of the filament of the bulb is focused on the screen.
- c) Measure and record the distance, a , between the lens and the screen.
- d) While keeping the distance, x , between the bulb and the screen fixed, move the lens towards the screen until a sharp diminished image of the filament of the bulb is focused on the screen.
- e) Measure and record the new distance, b , between the lens and the screen.
- f) Determine the distance, $y = (a - b)$.
- g) Repeat procedures (a) to (f) for values of $x = 90.0, 85.0, 80.0, 75.0$ and 70.0 cm.
- h) Tabulate your results including values of $(x + y)$, $(x - y)$ and $z = \frac{x}{(x - y)}$.
- i) Plot a graph of $(x + y)$ against z .
- j) Find the slope, S , of the graph.
- k) Calculate the focal length, f , of the lens from the expression

$$f = \frac{4d + S}{8}.$$

3. In this experiment, you will determine the electrical resistivity, ρ , of the material of the bare wire labelled W provided.

METHOD I

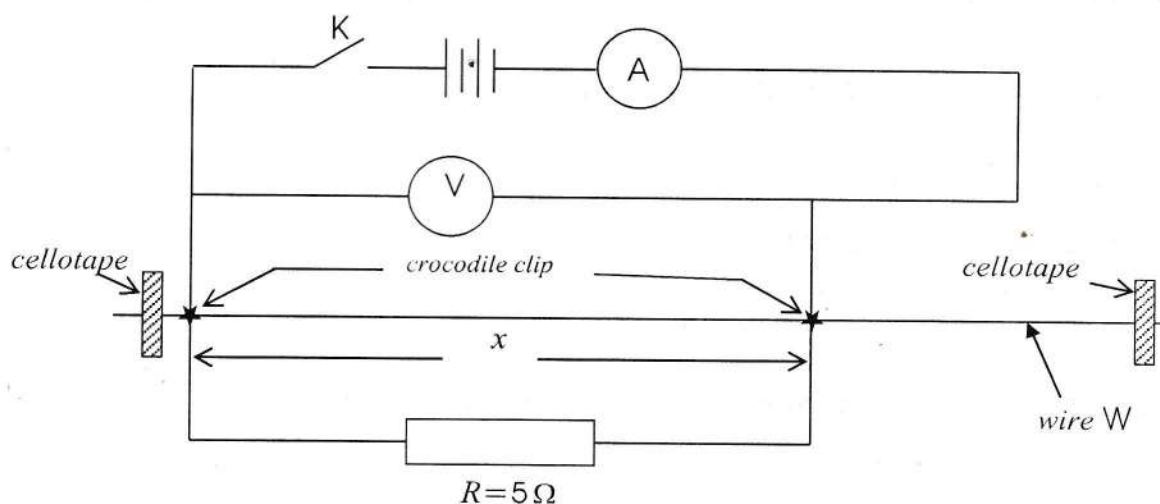


Fig. 3.1

- a) Measure and record the mean diameter, D , of the bare wire W.
- b) Set up the circuit in Figure 3.1 with length $x = 0.800$ m.
- c) Close switch K.
- d) Read and record the ammeter reading, I , and the voltmeter reading, V .

- e) Open switch K.
 f) Calculate β from the expression

$$\beta = 0.79D^2$$

 g) Calculate ρ_1 from the expression

$$\rho_1 = \frac{\beta VR}{x(IR - V)}$$

METHOD II

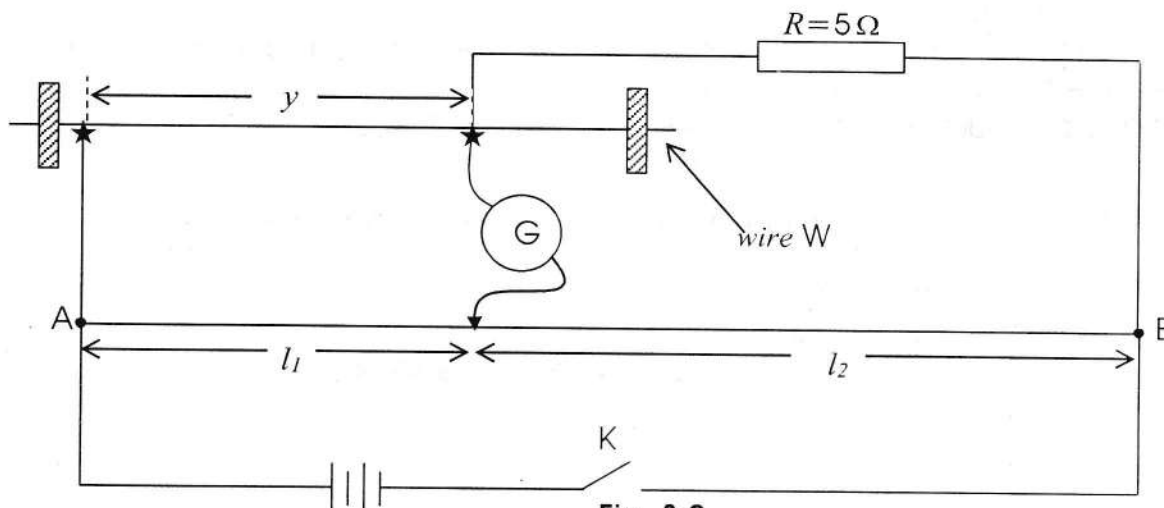


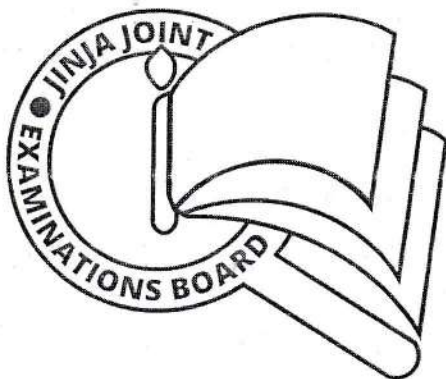
Fig. 3.2

- a) Connect the circuit in Figure 3.2 with a length, $y = 0.200$ m.
 b) Close the switch and move the sliding contact along the potentiometer wire AB until the galvanometer G shows no deflection.
 c) Measure and record the balance lengths l_1 and l_2 , in metres.
 d) Repeat procedures (a) to (c) for values of $y = 0.300, 0.400, 0.500, 0.600$ and 0.700 m.
 e) Enter all your results in a suitable table, including values of $\frac{l_1}{l_2}$ and $\frac{y}{R}$.
 f) Plot a graph of $\frac{y}{R}$ against $\frac{l_1}{l_2}$.
 g) Find the slope, S , of the graph.
 h) Calculate ρ_2 from the expression

$$\rho_2 = \frac{\beta}{S}$$

 i) Calculate the electrical resistivity, ρ , of the material of the bare wire W from the expression

$$\rho = \frac{\rho_1 + \rho_2}{2}$$



**JINJA JOINT EXAMINATIONS
BOARD**

**UGANDA ADVANCED
CERTIFICATE OF EDUCATION**

P510/3 PHYSICS PRACTICAL

**SAMPLE RESULTS
AND
MARKING GUIDE 2018**

Q.1

METHOD II

e) $h_0 = 0.376m$

g) $h_1 = 0.385m$

h) $y = 0.385 - 0.376$
 $= 0.009m$

j)

$m(kg)$	$h_1(m)$	$y(m)$	$\frac{1}{m}(kg^{-1})$	$\frac{1}{y}(m^{-1})$
0.150	0.385	0.009	6.67	111
0.200	0.389	0.013	5.00	77
0.250	0.392	0.016	4.00	63
0.300	0.396	0.020	3.33	50
0.350	0.399	0.023	2.86	43
0.400	0.403	0.027	2.50	37

l) $S = \frac{117 - 28}{7.2 - 2.0}$
 $= 17kgm^{-1}$

m) $E = \frac{9.81 \times 17 \times [(98.0 - 2.0) \times 10^{-2}]^3}{2.84 \times 10^{-8}}$
 $= 5.2 \times 10^9 Nm^{-2}$

n) $E = \frac{1}{2}(5.89 + 5.2) \times 10^9$
 $= 5.6 \times 10^9 Nm^{-2}$

Q.2

QUESTION 2

PART I

c) $d = 16.5\text{cm}$

PART II

c) $a = 73.5\text{cm}$

e) $b = 21.0\text{cm}$

f) $y = 73.5 - 21.0$
 $= 52.5\text{cm}$

h)

$x(\text{cm})$	$a(\text{cm})$	$b(\text{cm})$	$y(\text{cm})$	$(x + y)(\text{cm})$	$(x - y)(\text{cm})$	z
95.0	73.5	21.0	52.5	147.5	42.5	2.24
90.0	68.0	22.0	46.0	136.0	44.0	2.05
85.0	62.1	22.7	39.4	124.4	45.6	1.86
80.0	56.5	23.5	33.0	113.0	47.0	1.70
75.0	50.0	24.5	25.5	100.5	49.5	1.52
70.0	43.0	26.7	16.3	86.3	53.7	1.30

j) $S = \frac{166 - 70}{2.52 - 1.06}$
 $= 65.8\text{cm}$

k) $f = \frac{(4 \times 16.5) + 65.8}{8}$
 $= 16.5\text{cm}$

QUESTION 3

METHOD I

Q.3

a)
$$D = \frac{0.37 + 0.37 + 0.37}{3}$$

$$= 0.37 \text{ mm}$$

d)
$$I = 0.76 \text{ A}$$

$$V = 2.20 \text{ V}$$

f)
$$\beta = 0.79 \times (0.37 \times 10^{-3})$$

$$= 1.1 \times 10^{-7} \text{ m}^2$$

g)
$$\rho_1 = \frac{1.1 \times 10^{-7} \times 2.20 \times 5}{0.800 \times [(0.76 \times 5) - 2.20]}$$

$$= 9.5 \times 10^{-7} \Omega \text{ m}$$

METHOD II

c)
$$l_1 = 0.240 \text{ m} \quad l_2 = 0.760 \text{ m}$$

$$R = 5 \Omega$$

e)

$y(m)$	$l_1(m)$	$l_2(m)$	$\frac{l_1}{l_2}$	$\frac{y}{R} (m\Omega^{-1})$
0.200	0.240	0.760	0.32	0.040
0.300	0.324	0.676	0.48	0.060
0.400	0.372	0.628	0.59	0.080
0.500	0.422	0.578	0.73	0.100
0.600	0.477	0.523	0.91	0.120
0.700	0.506	0.494	1.02	0.140

g)
$$S = \frac{0.147 - 0.036}{1.09 - 0.29}$$

$$= 0.14 \text{ m}\Omega^{-1}$$

h)
$$\rho_2 = \frac{1.1 \times 10^{-7}}{0.14}$$

$$= 7.9 \times 10^{-7} \Omega \text{ m}$$

i)
$$\rho = \frac{(9.5 + 7.9) \times 10^{-7}}{2}$$

$$= 8.7 \times 10^{-7} \Omega \text{ m}$$

QUESTION 1 (40 marks)		
A ₁	Thickness t measured at least three times.....	$\frac{1}{2}$
A ₂	Value of $t = 6.20 - 7.00$, 2dp ; unit: mm (<i>correct symbols</i>).....	$\frac{1}{2} + \frac{1}{2}$
A ₃	Width w measured at least three times.....	$\frac{1}{2}$
A ₄	Value of $w = 2.30 - 2.70$, 2dp ; unit: cm (<i>correct symbols</i>).....	$\frac{1}{2} + \frac{1}{2}$
A ₅	Substitution in SI in $k=wt^3$	$\frac{1}{2}$
A ₆	Correctly calculated value of $k = (2.00 - 3.50) \times 10^{-8}$ [3sf $\times 10^{-8}$]; unit: m^4 (<i>correct symbols</i>)	$1 + \frac{1}{2}$
A ₇	Time for 20 oscillations (SC: 0 or 1dp , SW: 2dp); unit: s	$\frac{1}{2} + \frac{1}{2}$
A ₈	Correctly calculated value of T ; unit: s	$\frac{1}{2} + \frac{1}{2}$
	Stop Clock: 1dp if time for 20 oscillations is 0dp <10s 2dp if time for 20 oscillations is 0dp >10s or 1dp < 10s 3dp if time for 20 oscillations is 1dp >10s Stop Watch: 3dp if time for 20 oscillations is <10s 4dp if time for 20 oscillations is > 10s	
A ₉	Substitution in expression $E = 6.32 \times 10^2 \left(\frac{mx^3}{kT^2} \right)$	$\frac{1}{2}$
A ₁₀	Correctly calculated value of $E = (2.00 - 7.00) \times 10^9$ [3sf $\times 10^9$]; unit: Nm^{-2} (<i>correct symbols</i>)	$1 + \frac{1}{2}$
		09
B ₁	Value of h_0 , 3dp ; unit: m (<i>correct symbols</i>)	$\frac{1}{2} + \frac{1}{2}$
B ₂	Columnar table labelled: $m, h_1, y, \frac{1}{m}, \frac{1}{y}$ (<i>correct symbols</i>) @ $\frac{1}{2}$	$2 \frac{1}{2}$
B ₃	Units in brackets: $kg, m, m, kg^{-1}, m^{-1}$ (<i>correct symbols</i>) @ $\frac{1}{2}$	$2 \frac{1}{2}$
B ₄	Values of h_1 , 3dp increasing @ $\frac{1}{2}$	3
B ₅	Values of y correctly calculated, 3dp @ $\frac{1}{2}$	3
B ₆	Values of $\frac{1}{m}$ correctly calculated, 2dp @ $\frac{1}{2}$	3
B ₇	Values of $\frac{1}{y}$ correctly calculated, 0dp @ $\frac{1}{2}$	3
		18

C ₁	Title: A graph of $\frac{1}{y}$ against $\frac{1}{m}$, [no units, correct symbols]	$\frac{1}{2}$
C ₂	Axes perpendicular with arrows at end, correctly labelled; VA: $\frac{1}{y}(m^{-1})$ HA: $\frac{1}{m}(kg^{-1})$ [correct symbols] @ $\frac{1}{2}$	1
C ₃	Uniform scales, half page at least, starting values indicated, marked at least three times @ $\frac{1}{2}$	1
C ₄	Correctly plotted points [no shading] @ $\frac{1}{2}$	3
	[If axes not labelled or reversed, no mark]	
C ₅	Line of best fit provided 4 points correctly plotted	$\frac{1}{2}$
C ₆	Indication of slope covering all plotted points	$\frac{1}{2}$
	[If right angled triangle, must touch line of best fit]	
		<u>06 $\frac{1}{2}$</u>
D ₁	Coordinates for slope correctly read @axis $\frac{1}{2}$	$\frac{1}{2} + \frac{1}{2}$
D ₂	Correctly calculated $S = (15 - 19)$, 0dp ; unit: kgm^{-1}	$1 + \frac{1}{2}$
D ₃	Substitution in SI in expression for E	$\frac{1}{2}$
D ₄	Correctly calculated value of $E = (2.00 - 7.00) \times 10^9$ [3sf x 10^9]; unit: Nm^{-2} (correct symbols)	$1 + \frac{1}{2}$
D ₅	Determining average value of E	$\frac{1}{2}$
D ₆	Correctly calculated average value of $E = (2.00 - 7.00) \times 10^9$ [3sf x 10^9]; unit: Nm^{-2} (correct symbols)	$1 + \frac{1}{2}$
		<u>06 $\frac{1}{2}$</u>

QUESTION 2 (40 marks)

A ₁	Value of $d = (14.5 - 17.0)$, 1dp; unit: <i>cm</i>	2 + $\frac{1}{2}$
A ₂	Columnar table labelled: $x, a, b, y, (x + y), (x - y), z$ (<i>correct symbols</i>) @ $\frac{1}{2}$	3 $\frac{1}{2}$
A ₃	Units in brackets: <i>cm, cm, cm, cm, cm, cm, -</i> (<i>correct symbols</i>) @ $\frac{1}{2}$	3 $\frac{1}{2}$
A ₄	Values of $a = (75.0 - 40.0)$, decreasing, 1dp in <i>cm</i> or 3dp in <i>m</i> @ $\frac{1}{2}$	3
A ₅	Values of $b = (20.0 - 28.0)$, increasing, 1dp in <i>cm</i> or 3dp in <i>m</i> @ $\frac{1}{2}$	3
A ₆	Values of y correctly calculated, 1dp in <i>cm</i> or 3dp in <i>m</i> @ $\frac{1}{2}$	3
A ₇	Values of $(x + y)$ correctly calculated, 1dp in <i>cm</i> or 3dp in <i>m</i> @ $\frac{1}{2}$	3
A ₈	Values of $(x - y)$ correctly calculated, 1dp in <i>cm</i> or 3dp in <i>m</i> @ $\frac{1}{2}$	3
A ₉	Values of z correctly calculated, 2dp	3
		27 $\frac{1}{2}$
B ₁	Title: A graph of $(x + y)$ against z [<i>no units, correct symbols</i>]	$\frac{1}{2}$
B ₂	Axes perpendicular with arrows at end, correctly labelled; VA: $(x + y)(cm)$, HA: z [<i>correct symbols</i>] @ $\frac{1}{2}$	1
B ₃	Uniform scales, half page at least, starting values indicated, marked at least three times @ $\frac{1}{2}$	1
B ₄	Correctly plotted points [<i>no shading</i>] @ $\frac{1}{2}$	3
[If axes not labelled or reversed, no mark]		
B ₅	Line of best fit provided 4 points correctly plotted	$\frac{1}{2}$
B ₆	Indication of slope covering all plotted points	$\frac{1}{2}$
[If right angled triangle, must touch line of best fit]		
		06 $\frac{1}{2}$
C ₁	Coordinates for slope correctly read @ axis $\frac{1}{2}$	$\frac{1}{2} + \frac{1}{2}$
C ₂	Correctly calculated $S = (58.0 - 68.0)$, 1dp ; unit: <i>cm</i>	2 + $\frac{1}{2}$
C ₃	Correctly calculated value of $f = (14.5 - 17.0)$, 1dp ; unit: <i>cm</i>	2 + $\frac{1}{2}$
[provided correct substitution in expression $f = \frac{4d + S}{8}$, (<i>correct symbols</i>)]		06

QUESTION 3 (40 marks)

A ₁	Diameter D measured at least three times	½
A ₂	Value of D = 0.36 – 0.39, 2dp ; unit: <i>mm</i>	1+ ½
A ₃	Value of I = 0.50 – 1.00, 2dp ; unit: <i>A</i>	1+ ½
A ₄	Value of V = 2.00 – 2.80, 1dp or 2dp , unit: <i>V</i>	1+ ½
A ₅	Correctly calculated value of $\beta = 1.1$ or 1.2×10^{-7} , 1dp ; unit: m^2 [0.11 or 0.12, 2dp in mm^2]	1+ ½
A ₆	Correctly calculated value of $\rho_1 = (1.0 - 9.9) \times 10^{-7}$ [2sf $\times 10^{-7}$]; unit: Ωm ...	1+ ½
		08

B ₁	Columnar table labelled: $y, l_1, l_2, \frac{l_1}{l_2}, z$ (<i>correct symbols</i>) @½	2½
B ₂	Units in brackets: $m, m, m, -, m\Omega^{-1}$ (<i>correct symbols</i>) @½	2½
B ₃	Values of $l_1 = 0.200 - 0.600$; 3dp , <i>increasing</i> @1	6
B ₄	Values of $l_2 = 0.800 - 0.400$; 3dp , <i>decreasing</i> @1	3
B ₅	Values of $\frac{l_1}{l_2}$, correctly calculated, 2dp @½	3
B ₆	Values of $\frac{y}{R}$, correctly calculated, 3dp @½	3
		20

C ₁	Title: A graph of $\frac{y}{R}$ against $\frac{l_1}{l_2}$ [no units, correct symbols]	$\frac{1}{2}$
C ₂	Axes perpendicular with arrows at end, correctly labelled; VA: $\frac{y}{R}$ ($m\Omega^{-1}$), HA: $\frac{l_1}{l_2}$ [correct symbols] @	1
C ₃	Uniform scales, half page at least, starting values indicated, marked at least three times @ $\frac{1}{2}$	1
C ₄	Correctly plotted points [no shading] @ $\frac{1}{2}$ [If axes not labelled or reversed, no mark]	3
C ₅	Line of best fit provided 4 points correctly plotted	$\frac{1}{2}$
C ₆	Indication of slope covering all plotted points [If right angled triangle, must touch line of best fit]	$\frac{1}{2}$
		06 $\frac{1}{2}$
D ₁	Coordinates for slope correctly read @axis $\frac{1}{2}$	$\frac{1}{2} + \frac{1}{2}$
D ₂	Correctly calculated $S = (0.12 - 0.23)$, 2dp ; unit: $m\Omega^{-1}$	1 $\frac{1}{2}$
D ₃	Correctly calculated value of $\rho_2 = (1.0 - 9.9) \times 10^{-7}$ [2sf x 10^{-7}]; unit: Ωm (correct symbols)	1 $\frac{1}{2}$
D ₄	Correctly calculated value of $\rho = (1.0 - 9.9) \times 10^{-7}$ [2sf x 10^{-7}]; unit: Ωm (correct symbols)	1 $\frac{1}{2}$
		05 $\frac{1}{2}$
END.		

Candidate's Name

Signature

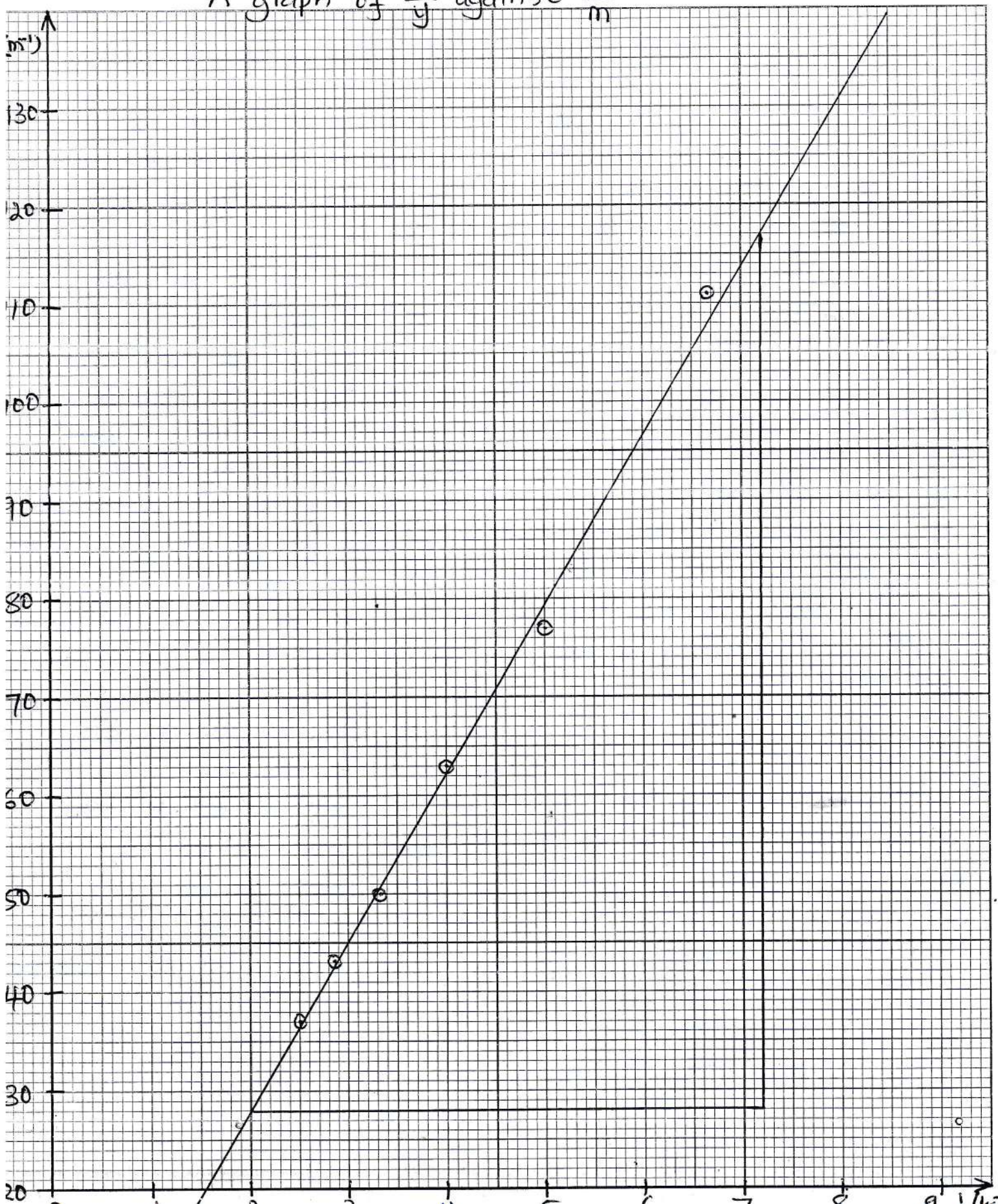
Subject Name

QUESTION ONE

Random No.

Personal Number

A graph of $\frac{1}{y}$ against $\frac{1}{m}$



Candidate's Name

Signature

QUESTION TWO

Random No.

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Personal Number

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Paper code /

Subject Name A graph of $(x+y)$ against z

